Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Invariant Bearing Fault Detection

Member 1, Member 2, Member 3, and Member 4

***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds do not transfer across load: the same absolute threshold either misses harmonics or floods the spectrum with crossings, depending on the operating point. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings carry the shaft in motors, pumps, turbines, gearboxes and many other machines. Localized surface damage can grow into a severe mechanical failure if it is not found early, so the condition of the bearing matters for machine availability. Each time the rolling contact passes a damaged spot on an inner race, an outer race or a rolling element, it leaves a short contact disturbance. Those impacts excite resonances in the bearing and in the housing, and an accelerometer near the machine records the ringing. The useful information is the *repetition rate* of the impacts, and demodulating the resonance recovers it. That is the principle behind envelope analysis [1], [2].

The measured vibration is not stationary, and load is the main reason. Load changes the contact force, and with it the strength of the defect-induced impacts; it also changes the overall vibration and the noise floor. The effective shaft speed differs between load conditions, and local slip pulls the true impact rate a little away from the value that the bearing geometry predicts. A fixed threshold on raw amplitude, or on the spectrum, therefore cannot be expected to work at every load.

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The approach here is pure DSP, not a model trained on labelled examples. Every stage can be inspected on its own, including the characteristic frequencies, the filters, the transforms and the detection thresholds. Nothing is hidden in learned weights. Interpretability does not guarantee generalization, though. Bearing geometry, shaft speed, sensor placement and validation data are still needed.

This paper applies two classical operations where they matter most. Kurtosis measures impulsiveness, so it finds the resonance that the defect excites [3], and that is how the filter band is chosen. The decision threshold is estimated *locally* in the envelope spectrum from an order statistic, which is not thrown off by the impulsive outliers that corrupt a moving average. The detector that comes out has no learned parameters and fits a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure inside a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with a calibrated Hann-windowed FFT, where squaring shifts the resonance to baseband by the modulation theorem and avoids the intermodulation that a square-root envelope produces;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, tested against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

A localized bearing defect behaves like a sequence of short impacts. Write the impact times as $t_m = m/f_{\text{fault}} + \varepsilon_m$, where f_{fault} is the nominal repetition rate and ε_m is a timing error from local slip or speed fluctuation. Impact m has amplitude A_m and excites the structure with impulse response $h(t)$. Background machine vibration and sensor noise enter as $w(t)$, so

$$x(t) = \sum_m A_m \, h(t - t_m) + w(t). \tag{1}$$

For an underdamped structural mode the causal impulse response is a decaying sinusoid,

$$h(t) = e^{-\sigma t} \cos(2\pi f_r t)\, u(t), \qquad t \geq 0, \tag{2}$$

where σ is the decay rate, f_r the resonance frequency and $u(t)$ the unit step. The response starts at the impact and decays afterwards, which is what makes it causal. The same repeated-impact, exponentially decaying picture is the classical localized-defect model [4].

Sampling with period $T_s = 1/f_s$ gives

$$x[n] = \sum_m A_m\, h[n - n_m] + w[n], \qquad n_m \approx \text{round}(f_s t_m). \quad (3)$$

The same signal can be written through the impact sequence $p[n] = \sum_m A_m \delta[n-n_m]$, as $x[n] = p[n] * h[n] + w[n]$, where $*$ is linear convolution. The nominal spacing between impacts is $N_p \approx f_s/f_{\text{fault}}$ samples. N_p need not be an integer, and small timing errors push the observed spectral peaks away from the ideal kinematic frequencies.

The sampling frequency has to be more than twice the highest vibration frequency that must be represented, $f_s > 2f_{\max}$. The acquisition chain also has to suppress out-of-band components before sampling, or aliasing folds them into the band of interest. The records used here run at 12 and 48 kHz (CWRU) and at 48.828 and 97.656 kHz (MFPT), which puts the Nyquist frequencies at 6, 24, 24.414 and 48.828 kHz.

In the z-plane, $h[n]$ gives a complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \quad (4)$$

and stability follows from $|z_{1,2}| < 1$. The model also makes the central point of the paper explicit. The fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r. Demodulation separates the two; direct spectral inspection does not.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) DC Offset Removal and Detrending: Accelerometers come with a DC offset. The sensor electronics and the signal conditioning hardware add a constant bias, and it appears as a large peak at 0 Hz. Its sidelobes then leak into the low-frequency bins that matter. Before any filtering, the signal $x[n]$ of length N is centered by subtracting its sample mean:

$$x_0[n] = x[n] - \mu_x, \quad \text{where} \quad \mu_x = \frac{1}{N}\sum_{m=0}^{N-1} x[m] \quad (5)$$

In the frequency domain this removes the $\omega = 0$ term of the DTFT:

$$X_0(e^{j0}) = \sum_{n=0}^{N-1} x_0[n] = 0 \quad (6)$$

After this step the filter bank sees only the vibration riding on a zero baseline. The subtraction is exact in the DFT sense: it changes the bin at $\omega = 0$ and leaves every other coefficient untouched.

2) Sub-Band Kurtosis Resonance Selection: Below about 1000 Hz the spectrum is crowded. Shaft harmonics, structural vibration and load-dependent friction noise all sit there, and they hide the bearing impulses. The impulses themselves are short impacts. Each time a rolling element hits a defect on the raceway, it excites a high-frequency resonance of the housing. That band carries sharp bursts. The low-frequency interference is stationary and does not. A resonance belongs to the structure, not to the fault: it rings at about the same frequency after every impact, which is why one band can collect many bursts at once.

A fixed bandpass filter cannot find the resonance reliably, because the frequency depends on the machine and on how it is mounted. The band has to be picked per record. A filter bank built on the Fast Kurtogram [3] does this without training data. The Nyquist interval $[0, f_s/2]$ is split at several dyadic and 1/3-binary levels. On a 12 kHz record the limit is 6000 Hz, so the dyadic levels give bands of 3000, 1500, 750 Hz and so on, while the 1/3-binary levels add widths of 2000, 1000, 500 Hz. The 2000 Hz band used in Section IV comes from that second set. At level k there are $M_k = \text{round}(2^k)$ bands, and band i has bandwidth and center frequency

$$\Delta f_k = \frac{f_s}{2M_k}, \quad f_{c,i} = \left(i + \frac{1}{2}\right)\Delta f_k \quad (7)$$

For each band output $y_{k,i}[n]$, impulsiveness is measured by the fourth standardized central moment, the Kurtosis [5]:

$$K(f_{c,i}, \Delta f_k) = \frac{\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^4}{\left(\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^2\right)^2} \quad (8)$$

with μ_y the mean of the band signal. The fourth power in the numerator is what makes spikes count: one large transient outweighs many small samples. The denominator, the squared variance σ_y^4, normalizes out the band energy, so a loud band does not win by being loud. Gaussian noise gives $K \approx 3$. A band full of repeated impacts gives K well above 3. The band with the largest K is selected:

$$\{f_c^*, BW^*\} = \arg\max_{f_{c,i},\,\Delta f_k} K(f_{c,i}, \Delta f_k) \quad (9)$$

Its cutoffs are $f_{\text{low}} = f_c^* - BW^*/2$ and $f_{\text{high}} = f_c^* + BW^*/2$, that is $\omega_1 = 2\pi f_{\text{low}}/f_s$ and $\omega_2 = 2\pi f_{\text{high}}/f_s$.

3) Bandpass Filter Design: Kaiser FIR versus Butterworth IIR: Two filters are designed for that passband, a linear-phase FIR and a Butterworth IIR. Both have to reject out-of-band motor interference and broadband noise.

The FIR is Type-I, with even order M and $L = M + 1$ taps. The ideal bandpass impulse response is truncated around its symmetry center $\alpha = M/2$ and multiplied by a Kaiser window [6]:

$$h_{\text{FIR}}[n] = \left(\frac{\sin(\omega_2(n-\alpha)) - \sin(\omega_1(n-\alpha))}{\pi(n-\alpha)}\right) \times \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n-\alpha}{\alpha}\right)^2}\right)}{I_0(\beta)} \quad (10)$$

for $0 \le n \le M$. $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and β sets the trade between transition width and sidelobe attenuation. The taps are symmetric, $h_{\text{FIR}}[n] = h_{\text{FIR}}[M-n]$, so the phase is exactly linear and the delay is flat:

$$\tau_{g,\text{FIR}}(\omega) = -\frac{d\angle H_{\text{FIR}}(e^{j\omega})}{d\omega} = \frac{M}{2} \text{ samples} \quad (11)$$

An FIR of this form has all its poles at $z = 0$ and only zeros elsewhere, so it is stable whatever the coefficients do. The price is arithmetic: a longer window buys a steeper transition, and every extra tap costs one more operation per sample. The design in Section IV uses $\beta = 8$, which targets roughly -80 dB of stopband attenuation.

The second design is a 4th-order Butterworth bandpass, obtained with the bilinear transform [7]. That transform maps the left half of the s-plane into the unit circle:

$$s = \frac{2}{T_s}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) \tag{12}$$

with $T_s = 1/f_s$. The mapping compresses frequency non-linearly, so the digital cutoffs are pre-warped before the prototype is transformed:

$$\Omega_k = \frac{2}{T_s}\tan\!\left(\frac{\omega_k}{2}\right), \quad k \in \{1, 2\} \tag{13}$$

After pre-warping, the -3 dB points land exactly on f_{low} and f_{high}, and the passband stays maximally flat.

4) Cascade SOS Realization and Zero-Phase Filtering: A 4th-order bandpass has eight poles. Written in direct form, that is a degree-eight denominator, and in fixed point a small error in one coefficient moves the poles a long way. They can cross the unit circle. The fix is to factor the transfer function into $K = 4$ second-order sections:

$$H_{\text{IIR}}(z) = G\prod_{k=1}^{K} H_k(z) = G\prod_{k=1}^{K}\frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \tag{14}$$

where G is the overall gain. Each section holds one conjugate pair $p_k, p_k^* = r_k e^{\pm j\theta_k}$. Stability needs the ROC $|z| > r_{\max}$ to contain the unit circle, so every pole radius must satisfy

$$r_k = \sqrt{a_{2k}} < 1, \quad \forall k \in \{1, 2, \ldots, K\} \tag{15}$$

The cascade is cheap to run, but its phase is not linear near the band edges. Group delay then varies with frequency, which smears a short impact in time and lowers its peak. Forward-backward filtering removes the problem. The record is filtered, reversed, filtered again, and reversed back:

$$Y_{\text{BP}}(z) = H(z)H(z^{-1})X_0(z) \tag{16}$$

On the unit circle, $H(e^{-j\omega}) = H^*(e^{j\omega})$ for real coefficients, so

$$H_{\text{zero-phase}}(e^{j\omega}) = H(e^{j\omega})H^*(e^{j\omega}) = |H(e^{j\omega})|^2 \tag{17}$$

That response is real and non-negative, so the phase is zero and the group delay is zero samples. The impacts keep their time positions and their shape, and the stopband attenuation doubles in decibels. The passband edge shifts a little inward, because the effective filter is the square of the one that was designed.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

The bandpass stage leaves $y_{\text{BP}}[n]$ as a series of short bursts of structural ringing. The bursts repeat at the fault rate f_{fault}. The FFT of $y_{\text{BP}}[n]$ puts the energy around the resonance ω_r, so the fault rate is almost invisible in the spectrum. It has to be moved down to a low-frequency region, where the harmonics are far apart and easy to separate. That operation is demodulation.

The first step is the analytic signal, built with the Discrete Hilbert Transform [6]:

$$z[n] = y_{\text{BP}}[n] + j\hat{y}_{\text{BP}}[n], \tag{18}$$

where $\hat{y}_{\text{BP}}[n] = \mathcal{H}\{y_{\text{BP}}[n]\}$ is the Hilbert transform of the filtered signal. A real record has a spectrum on both sides of 0 Hz, and the two halves mirror each other. The transform keeps the positive half and drops the rest, which is what makes the envelope well defined. It works as a 90° phase shifter,

$$H_{\text{Hilbert}}(e^{j\omega}) = -j\ \operatorname{sgn}(\omega), \tag{19}$$

so every positive frequency is shifted by $-\pi/2$ and the negative frequencies are removed.

The usual envelope takes a square root of $y_{\text{BP}}^2 + \hat{y}_{\text{BP}}^2$. The squared envelope is used here instead:

$$e[n] = |z[n]|^2 = z[n]z^*[n] = y_{\text{BP}}^2[n] + \hat{y}_{\text{BP}}^2[n]. \tag{20}$$

Squaring does two useful things. A square in time is an auto-convolution in frequency, so the modulation theorem moves the carrier ω_r down to baseband (0 Hz) without distortion. A square root does not behave this way. It is non-linear, and the intermodulation harmonics it adds can hide a weak fault. Squaring also favours large amplitudes: an impact peak is squared into a much larger number, while the background changes little. The fault-to-noise ratio therefore improves before any threshold is applied.

The squared envelope is never negative, so it carries a large DC term. That term is much bigger than the fault harmonics, and its mainlobe leaks into the low-frequency bins where the cage and shaft harmonics sit. The mean is removed first:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N}\sum_{n=0}^{N-1} e[n]. \tag{21}$$

A fault frequency rarely sits exactly on a DFT bin centre. At 107.36 Hz, with $\Delta f = f_s/N_{\text{FFT}}$, the mismatch is unavoidable. Rectangular truncation then spreads sidelobes into the neighbouring bins. An N-point Hann window controls them:

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1. \tag{22}$$

The window brings both ends to zero, which halves the coherent amplitude. It also widens the mainlobe, so a little frequency resolution is lost, and that trade is accepted for the much lower sidelobes. The coherent gain

$$CG = \frac{1}{N}\sum_{n=0}^{N-1} w[n] = \tfrac{1}{2} \tag{23}$$

puts the amplitude back, so the SES levels keep their physical meaning. The one-sided SES is then computed with an N_{FFT}-point radix-2 FFT:

$$\text{SES}[k] = \frac{2}{N \cdot CG}\left|\sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\text{FFT}}}kn}\right|, \tag{24}$$

for $k = 0, 1, \ldots, N_{\text{FFT}}/2$, where bin k corresponds to $f_k = k\, f_s/N_{\text{FFT}}$ Hz.

D. Local Noise-Floor Adaptive Spectral Thresholding

Load changes the SES floor in both directions. At 0 HP the rolling elements press lightly on the raceways. The impacts are weak, and a fixed threshold set for normal running sits above the fault peaks, so they are missed. At 3 HP friction raises the whole floor, and the same threshold is crossed by noise in dozens of bins. No single fixed value satisfies both cases.

A CFAR rule [8] handles both cases. It slides a window along the SES and estimates the local floor from the bins around the test bin, so the comparison is always made against the floor that is actually there. No training data is needed, and the false-alarm rate stays fixed as the load changes.

For a test bin k, the N_G bins on each side are kept out of the estimate. The Hann mainlobe is four bins wide, so $N_G = 4$ is enough to stop a fault peak from entering its own estimate. The $N_R = 25$ bins beyond each guard band form the reference window

$$\mathcal{W}_k = \big\{\mathrm{SES}[m] \;\big|\; m \in [k - N_G - N_R,\, k - N_G - 1] \cup [k + N_G + 1,\, k + N_G + N_R]\big\}. \tag{25}$$

Two estimators are compared on $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Cell averaging takes the mean of the window, as a moving-average filter would:

$$\hat{\mu}[k] = \frac{1}{2N_R} \sum_{x \in \mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\, \hat{\mu}[k], \tag{26}$$

with $\alpha = 3.5$. This works while the floor changes slowly. It fails when a harmonic enters the window. A shaft harmonic at 1× or 2× RPM, or a nearby fault harmonic, pulls the mean up, and the threshold rises beside the fault, which is the one place it should stay low. A mean is easy to move: one large bin is enough. This is harmonic target masking, and it is the reason the next estimator exists.

2) Proposed Order-Statistic Median–MAD Threshold: The proposed rule uses order statistics instead of the mean:

$$\begin{aligned} \hat{m}[k] &= \mathrm{median}\,(\mathcal{W}_k)\,, \\ \mathrm{MAD}[k] &= \mathrm{median}\,(|\mathcal{W}_k - \hat{m}[k]|)\,. \end{aligned} \tag{27}$$

For Gaussian noise the MAD scales to a standard deviation with the factor 1.4826, the value that makes the two agree for a normal distribution,

$$\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k], \tag{28}$$

and the threshold is

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k])\,, \tag{29}$$

with $\gamma = 4.5$. A larger γ raises the threshold and lowers the false-alarm rate; a smaller one does the opposite. The median has a breakdown point of 50% [9], so up to half of the window can be contaminated before T_{MAD} moves. The threshold follows the floor from 0 HP to 3 HP.

3) Slip-Tolerant Harmonic Comb: Heavy radial load also causes slip. The rolling elements skid by 1–2%, so the measured fault frequency is not the kinematic value. The detector therefore searches a band, not a single bin,

$$\mathcal{B}_h = \left[h f_{\mathrm{fault}}(1 - \delta_{\mathrm{slip}}),\; h f_{\mathrm{fault}}(1 + \delta_{\mathrm{slip}})\right], \tag{30}$$

TABLE I
IDEAL KINEMATIC FAULT FREQUENCIES CALCULATED FROM THE SPECIFIED BEARING GEOMETRIES AND SHAFT SPEEDS

Condition	Speed (rpm)	BPFO	BPFI	BSF	FTF
		(Hz)			
CWRU 0 HP	1797	107.36	162.19	70.58	11.93
CWRU 1 HP	1772	105.87	159.93	69.60	11.76
CWRU 2 HP	1750	104.56	157.94	68.74	11.62
CWRU 3 HP	1730	103.36	156.14	67.95	11.48
MFPT (25 Hz shaft)	1500	81.12	118.88	63.86	10.14

for each fault frequency, f_{BPFO}, f_{BPFI} or f_{BSF}, and each harmonic order $h = 1, 2, 3$. The half-width is $\delta_{\mathrm{slip}} = 1.5\%$, narrow enough to exclude the neighbouring harmonic orders and wide enough to cover the slip. A harmonic is present when

$$\max_{f_k \in \mathcal{B}_h} (\mathrm{SES}[k] - T_{\mathrm{MAD}}[k]) > 0. \tag{31}$$

III. Experimental Setup and Datasets

A. Bearing Kinematics

Let $f_r = \mathrm{RPM}/60$ be the shaft frequency, n_b the number of rolling elements, d and D the element and pitch diameters and α the contact angle. With the diameter ratio $\rho = (d/D)\cos\alpha$, the ideal characteristic frequencies are

$$\begin{aligned} f_{\mathrm{BPFO}} &= \frac{n_b}{2} f_r\,(1 - \rho)\,, & f_{\mathrm{BPFI}} &= \frac{n_b}{2} f_r\,(1 + \rho)\,, \\ f_{\mathrm{BSF}} &= \frac{D}{2d} f_r \left(1 - \rho^2\right), & f_{\mathrm{FTF}} &= \frac{f_r}{2}\,(1 - \rho)\,, \end{aligned} \tag{32}$$

The CWRU drive-end bearing is an SKF 6205-2RS with nine rolling elements, a 7.94 mm element diameter, a 39.04 mm pitch diameter and a 0° contact angle. Its calculated frequencies at 1797 rpm agree with published CWRU benchmark data [10]. The MFPT bearing uses the NICE geometry documented for that dataset: eight rolling elements, 0.235 in element diameter, 1.245 in pitch diameter and 0° contact angle. Geometry goes into the calculator explicitly, so each dataset gets the right bearing.

Table I lists the CWRU frequencies at the four load settings, and the MFPT frequencies at the stated shaft rate of 25 Hz (1500 rpm). All of them are ideal targets from geometry. Rolling-element slip and speed variation can move the real fault peaks, and the detector of Section II-D is built to allow for that.

B. Datasets

CWRU: the Case Western Reserve University Bearing Data Center [11] provides accelerometer recordings for normal bearings and for seeded inner-race, outer-race and rolling-element defects. A public NumPy transcription of the archive [12] keeps the time series in `.npz` files, with channel keys such as DE, FE and BA. The file name carries the shaft-speed family, the defect type, the defect size and, for many fault files, the drive-end sampling rate in a `DE12` or `DE48` suffix. The loader reads all of it from that name: RPM, fault label, defect diameter, race position and the rate suffix.

Two cases need care. Normal DE files have no rate suffix, and the BA rate is never inferred, so both need a sampling rate that has been verified rather than a silent default. FE channels run at 12 kHz even when they sit beside `DE48` data. The load conditions used here are 1797 rpm (0 HP), 1772 rpm (1 HP), 1750 rpm (2 HP) and 1730 rpm (3 HP). The wider dataset description sometimes quotes a speed range down to about 1720 rpm, so an individual recording takes its speed from its metadata where that exists, not from the load label [10].

MFPT: the standard MATLAB recordings hold a vibration vector `gs` together with load, shaft-rate and sample-rate metadata [13]. The varied-load outer-race records cover 25, 50, 100, 150, 200, 250 and 300 lb; the varied-load inner-race records cover 0, 50, 100, 150, 200, 250 and 300 lb. Baseline recordings sit at 270 lb and 97.656 kHz for 6 s, and fixed-load outer-race records use 270 lb at the same high rate. The varied-load fault records run at 48.828 kHz for 3 s.

C. Signal Preprocessing

The first step subtracts the sample mean,

$$x_0[n] = x[n] - \frac{1}{N}\sum_{n=0}^{N-1} x[n], \tag{33}$$

A constant DC offset would otherwise put energy at zero frequency. Optional peak normalization divides $x_0[n]$ by $\max|x_0[n]|$; RMS normalization divides by $\sqrt{(1/N)\sum_n x_0^2[n]}$. The demeaned signal stays as the reference in the implementation. Normalization is for comparing waveform shape only. It should not be used to flatten real load-dependent amplitude differences in quantitative comparisons.

D. Baseline Spectra and Window Comparison

The baseline FFT runs after mean removal and optional windowing. For a window $w[n]$ the windowed transform is

$$X_w[k] = \sum_{n=0}^{N-1} x_0[n]\, w[n]\, e^{-j2\pi kn/N}. \tag{34}$$

A coherent-gain-corrected one-sided amplitude spectrum comes from dividing the magnitude by $\sum_n w[n]$ and doubling the positive-frequency bins other than DC and, for even N, the Nyquist bin. The STFT applies the same transform to successive overlapping frames, so frequency content can be followed in time.

A finite record may hold a non-integer number of cycles of a sinusoid. Multiplication by the observation window then spreads energy into neighbouring DFT bins, which is spectral leakage. A rectangular window has a narrow main lobe and comparatively high sidelobes. Hann and Blackman windows lower the sidelobes and pay for it with a wider main lobe. The implementation compares all three on the same input and sampling settings, first on a controlled off-bin sinusoid and then on the STFT of a record.

The controlled test uses $f_s = 12$ kHz, $N = 2048$ and an off-bin tone at 1467.01 Hz. Table II gives the fraction of FFT energy that falls outside an approximate, window-specific main-lobe region. It is an educational comparison, not a universal standard.

TABLE II
APPROXIMATE LEAKAGE-ENERGY FRACTION FOR A CONTROLLED OFF-BIN SINUSOID

Window	Leakage-energy fraction
Rectangular	0.06613550
Hann	0.00017592
Blackman	0.00000295

E. Pipeline Parameters

Unless stated otherwise, the experiments use $f_s = 12$ kHz, a 4th order Butterworth bandpass in cascade SOS form with forward–backward filtering, a Hann-windowed FFT with coherent-gain calibration, guard bins $N_G = 4$, $N_R = 25$ reference bins on each side, $\alpha = 3.5$ for CA-CFAR and $\gamma = 4.5$ for the Median–MAD rule. The SES uses the next power of two above the frame length (32 768 bins for a 24 000-sample frame, $\Delta f \approx 0.37$ Hz), while the baseline spectra use the full record length ($\Delta f \approx 0.10$ Hz for a 121 991-sample CWRU record). The static threshold stays at $T_{\text{static}} = 0.055$ on every record, so any change in its performance across loads comes from the load itself.

IV. Results and Discussion

A. Evaluation Metrics

Four measures are computed on the SES: spectral SNR, true detection rate, false-alarm rate and execution latency. Each is scored on every record at every load and defect size listed in Section III. Every CWRU record is split into five non-overlapping 2.0 s segments ($N = 24\,000$ samples at 12 kHz), and each segment is processed independently; a reported number is the mean over those segments, with the sample standard deviation where stated. The MFPT records last 3 s, so each contributes one 2.0 s segment.

The data come from the CWRU 2 HP Reliance Electric rig with a dynamometer load and the 6205-2RS drive-end bearing, sampled at 12 kHz over the four load steps of Table I, with outer-race spalls of 7, 14 and 21 mil. The MFPT records add seven radial load levels from 25 to 300 lbs. The fault frequencies of Table I, $f_{\text{BPFO}} = 107.36$ Hz and $f_{\text{BPFI}} = 162.19$ Hz at no load, serve as ground truth.

1) Spectral SNR. The first measure sets the energy at the harmonics against the local noise floor,

$$\text{SNR}_{\text{SES}} = 10\log_{10}\frac{\sum_{m=1}^{M}|\text{SES}[k_m]|^2}{\frac{1}{|\mathcal{K}_{\text{noise}}|}\sum_{k\in\mathcal{K}_{\text{noise}}}|\text{SES}[k]|^2}, \tag{35}$$

where k_m runs over the bins inside the ±1.5% slip window $\mathcal{B}_m$ around each harmonic $m f_{\text{fault}}$, and $\mathcal{K}_{\text{noise}}$ collects the non-harmonic bins of the 10–400 Hz demodulation band. The difference $\Delta\text{SNR} = \text{SNR}_{\text{SES}} - \text{SNR}_{\text{raw}}$ is the gain the filter and envelope stages add.

2) True detection rate. The second measure is the share of the targeted harmonics (1×, 2× and 3 × f_{fault}) that clear the threshold,

$$\text{TDR} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\Big(\max_{k\in\mathcal{B}_m}\text{SES}[k] > T[k]\Big)\times 100\%, \tag{36}$$

with $\mathbb{I}(\cdot)$ the indicator function and $\mathcal{B}_m = [m f_{\text{fault}}(1 - 0.015),\ m f_{\text{fault}}(1 + 0.015)]$ the slip search boundary.

3) False-alarm rate. The third measure is the share of the eligible bins above 15 Hz, outside the harmonic masks, that still cross $T[k]$,

$$\text{FAR} = \frac{100\%}{|\mathcal{K}_N|} \sum_{k \in \mathcal{K}_N} \mathbb{I}(\text{SES}[k] > T[k]), \tag{37}$$

where $\mathcal{K}_N$ collects the non-harmonic bins above 15 Hz. A rate keeps records with different FFT lengths comparable.

4) Execution latency. The last measure is the wall-clock time for one 2.0 s frame ($N = 24\,000$ samples at 12 kHz), taken on the same machine for every configuration.

B. Filter Architecture and Pole–Zero Stability Analysis

The Kurtogram was run on the first 2.0 s segment of the 0 HP outer-race record (7 mil defect). Over the levels spanning $[0, 6000]$ Hz it chose a band centered at $f_c^* = 5181.8$ Hz with $BW^* = 545.5$ Hz, which puts the cutoffs at 4909.1 Hz and 5454.5 Hz. Inside that band the kurtosis peaked at $K = 7.81$. The broadband input sat at $K = 7.56$.

With the band fixed, a Type-I Kaiser FIR ($M = 128$, $L = 129$ taps, $\beta = 8.0$) was compared against a 4th-order Butterworth IIR in cascade SOS form (four biquads). Fig. 1 shows the pole–zero map, the magnitude response, the phase response and the group delay.

In Fig. 1(a) the FIR has zeros spread along the stopband arcs and every pole at the origin. The IIR is different. The bilinear transform puts four zeros at $z = 1$, which blocks DC, and four at $z = -1$, which blocks Nyquist, with four conjugate pole pairs inside the passband arc. Its largest pole radius is $r_{\max} = 0.9626$. The region of convergence $|z| > 0.9626$ contains the unit circle, so the filter is BIBO stable. The four zeros at $z = \pm 1$ are exact nulls rather than deep attenuation, which is why DC and Nyquist are blocked completely.

Fig. 1(b) compares magnitudes. Both filters are flat at 0 dB across $[4909, 5455]$ Hz, and both reject the motor region below 1000 Hz by more than 100 dB. The 129-tap FIR rolls off faster and reaches below -110 dB, but it costs 129 multiply-accumulate operations per sample. The four-section SOS filter needs $4 \times 5 = 20$, which is 84% fewer.

Fig. 1(c) is the unwrapped phase. The FIR phase is exactly linear, the signature of a symmetric Type-I design, while the IIR phase bends through the passband because its poles and zeros are not phase-matched. Fig. 1(d) is the group delay. In causal operation the FIR delays every frequency by the same 64 samples. The IIR does not: its delay rises from about 20 samples at the lower cutoff to about 38 samples at 5455 Hz, where the poles come close to the unit circle. With forward–

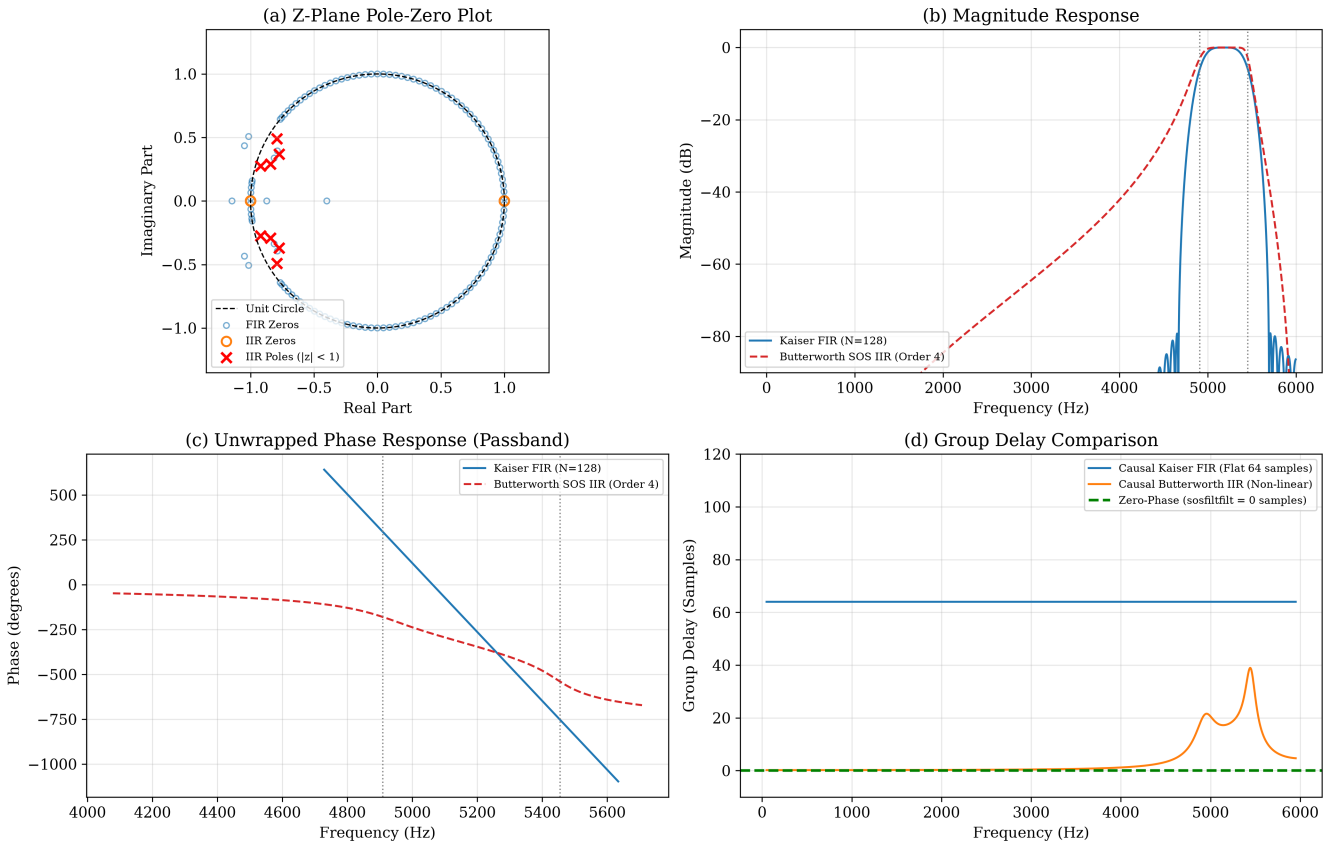


Fig. 1. Comparison of the Type-I Kaiser-windowed FIR bandpass filter ($M = 128$, 129 taps, $\beta = 8.0$) and the 4th-order Butterworth Cascade SOS IIR bandpass filter across the Kurtosis-selected passband $[4909, 5455]$ Hz: (a) complex z-plane pole–zero plot verifying that all IIR poles lie strictly inside the unit circle ($r_{\max} = 0.9626 < 1$), (b) magnitude response $|H(e^{j\omega})|$ in decibels, (c) unwrapped phase response in the passband, and (d) group delay comparison showing constant causal FIR delay (64 samples), non-linear causal IIR band-edge delay, and 0 samples net delay under zero-phase forward–backward filtering.

backward filtering the effective response is $|H(e^{j\omega})|^2$, the phase term cancels, and the delay is zero samples everywhere.

C. Time-Domain Resonance Demodulation Progression

Fig. 2 follows one segment through three stages, over 100 ms (1200 samples at 12 kHz).

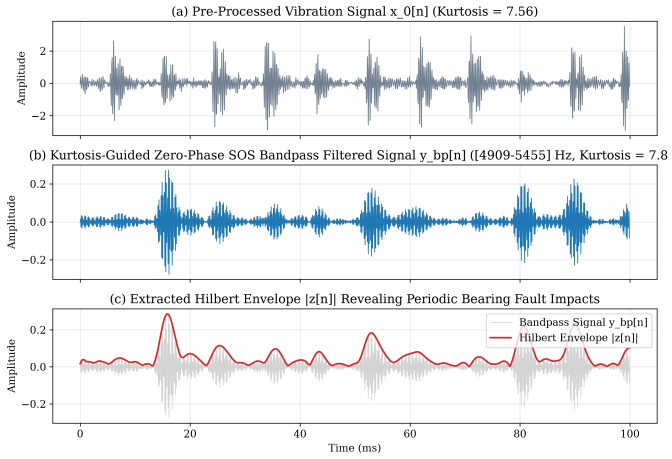


Fig. 2. Time-domain signal progression across a 100 ms segment of the 0 HP outer-race record: (a) pre-processed zero-mean vibration signal $x_0[n]$ with periodic impact bursts ($K = 7.56$), (b) Kurtosis-guided zero-phase SOS bandpass filtered output $y_{\mathrm{BP}}[n]$ across $[4909, 5455]$ Hz isolating the resonance that carries the impacts ($K = 7.81$), and (c) extracted Hilbert envelope $|z[n]|$ overlaid on $y_{\mathrm{BP}}[n]$, revealing the periodic bearing fault impulses.

In Fig. 2(a) the zero-mean record already shows the impact train. At 107.36 Hz the 100 ms window holds about eleven bursts, each an underdamped ringing of the structural resonance, spaced by roughly 9.3 ms. The signal is impulsive: its kurtosis is 7.56, well above the Gaussian value of 3.0, because the impacts populate the tails of the amplitude distribution.

In Fig. 2(b) the signal has passed through the Kurtosis-guided zero-phase SOS filter for $[4909, 5455]$ Hz. The low-frequency shaft and motor content is removed and the resonance bursts remain, one per impact on the outer race. The kurtosis moves from 7.56 to 7.81: the band concentrates the impulsive energy, but the raw record is already impulsive, so this statistic changes little. The band choice still matters for the envelope, because it removes the low-frequency floor that would otherwise dominate the demodulated spectrum.

Fig. 2(c) overlays the Hilbert envelope $|z[n]|$ on the filtered waveform. The envelope follows the top of each burst, so the 5.2 kHz carrier becomes a sequence of positive pulses. Their spacing is $\Delta t \approx 9.31$ ms, which is $1/107.36$ Hz, the outer-race fault frequency. The spacing of the envelope peaks, not their height, carries the fault information here. That envelope is what the SES stage consumes.

D. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. The fixed threshold is not scale-invariant: at 0.055 it lies above the whole spectrum in all four panels, so no BPFO harmonic crosses it and the static rule marks nothing. Across the five benchmark segments it fires only in one 0 HP segment that contains an unusually large transient, which is why its detection rate in Table III is not zero. The adaptive rules behave differently. Both CA-CFAR and Median–MAD follow the load-dependent floor, and both flag the 1×, 2× and 3× BPFO harmonics in every panel, so the harmonic comb survives the load change once the threshold tracks the noise. The two adaptive rules differ quantitatively rather than qualitatively; Table III measures that difference.

E. Quantitative Benchmark and Comparative Discussion

Table III applies the three threshold rules to the same Kurtosis-guided SES, so the detector is the only thing that changes between rows. Fig. 3 shows the spectra behind the outer-race numbers.

1) Static threshold. The fixed level of 0.055 was set in dataset units, and it does not transfer. At 1, 2 and 3 HP no bin crosses it. At 0 HP four of the five segments also stay below it; the 20% mean comes from a single segment whose transient pushes the harmonic peaks over the line. A fixed threshold therefore reports neither a stable detection rate nor a stable false-alarm rate across the load sweep.

2) CA-CFAR. Cell averaging follows the load-dependent floor, and its false-alarm share stays between 5.2% and 7.0% at every load. Its detection rate moves with the segment: 100% at 0 HP, 60% at 1 HP, 86.7% at 2 HP and 100% at 3 HP. The large standard deviations at 1 and 2 HP (54.8 and 29.8 points) show that some segments lose the higher harmonics while others keep them, because the reference window inherits energy from the fault lines themselves.

3) Median–MAD. The proposed rule returns exactly the same detection rate as CA-CFAR at every load, segment for segment, but its false-alarm share is higher: 10.3% against 7.0% at 0 HP, 7.4% against 5.8% at 1 HP, 6.4% against 5.2% at 2 HP and 7.9% against 6.1% at 3 HP. The median and the MAD are robust to contaminated reference windows, but at $\gamma = 4.5$ the resulting threshold sits below the cell-averaged one, so the robustness is paid for in crossings. The operating point γ is the parameter that trades those crossings against detection margin.

4) SNR gain. The filter and envelope stages add between 19.2 and 26.7 dB of spectral SNR at the fault harmonics, and the gain is largest at full load, where the raw record sits at −2.06 dB and the SES reaches +24.60 dB. The detector choice does not affect this column, because all three rules read the same SES.

F. Finite-Word-Length and Quantization Error Analysis

On a microcontroller or a fixed-point DSP every value carries a limited number of bits, so the rounding along the chain has to be checked. The design was evaluated in a 16-bit Q1.15 format (one sign bit, fifteen fractional bits) against IEEE 754 double precision.

1) Filter structure and stability. Direct-form IIR filters are sensitive to coefficient rounding. A pole can land on the unit circle or outside it. The cascade SOS form of Section II-B

Fig. 3: Multi-Load Squared Envelope Spectrum (SES) with Adaptive Spectral Thresholding

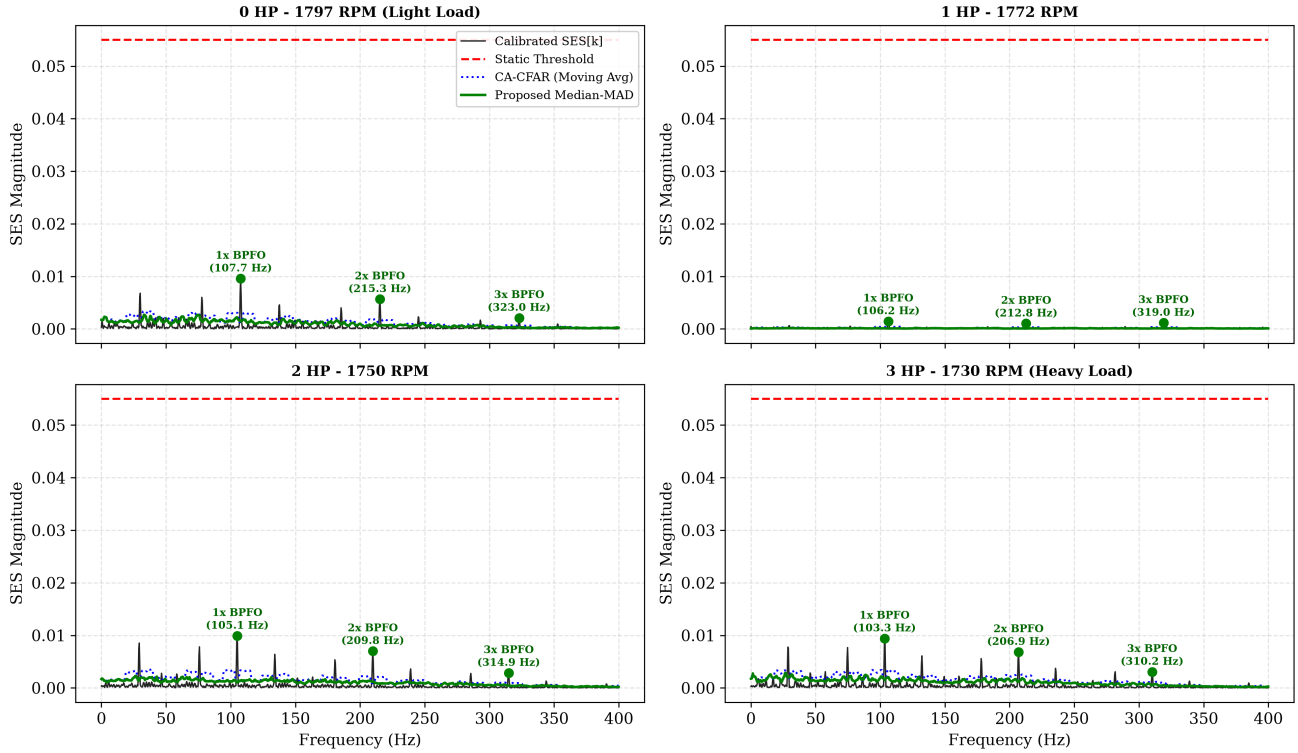


Fig. 3. Calibrated SES and comparative thresholding at four motor loads for the outer-race 7 mil record. The static threshold of 0.055 sits above the SES at every load and marks no harmonic; CA-CFAR and Median–MAD follow the load-dependent floor and flag all three BPFO harmonics in every panel.

TABLE III

QUANTITATIVE PERFORMANCE BENCHMARKING ACROSS OPERATIONAL LOADS AND DETECTORS FOR THE OUTER-RACE 7 MIL RECORD; MEAN ± SAMPLE STANDARD DEVIATION OVER THE FIVE 2.0 S SEGMENTS OF EACH RECORD. THE STATIC RULE USES $T_{\text{STATIC}} = 0.055$, CA-CFAR USES $\alpha = 3.5$ AND THE PROPOSED RULE USES $\gamma = 4.5$. FAR IS THE PERCENTAGE OF ELIGIBLE NON-HARMONIC BINS ABOVE THE THRESHOLD.

Load	Detector	TDR (%)	FAR (%)	Raw SNR (dB)	SES SNR (dB)	ΔSNR (dB)
0 HP (0 N·m)	Static (0.055)	20.0 ± 44.7	1.4 ± 3.2	+1.95	+27.40	+25.45 ± 2.72
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	7.0 ± 2.1	+1.95	+27.40	+25.45 ± 2.72
	Median–MAD	100.0 ± 0.0	10.3 ± 3.7	+1.95	+27.40	+25.45 ± 2.72
1 HP (2.4 N·m)	Static (0.055)	0.0 ± 0.0	0.0 ± 0.0	−0.21	+18.99	+19.20 ± 8.71
	CA-CFAR ($\alpha = 3.5$)	60.0 ± 54.8	5.8 ± 2.8	−0.21	+18.99	+19.20 ± 8.71
	Median–MAD	60.0 ± 54.8	7.4 ± 2.5	−0.21	+18.99	+19.20 ± 8.71
2 HP (4.8 N·m)	Static (0.055)	0.0 ± 0.0	0.0 ± 0.0	+0.86	+21.20	+20.35 ± 5.38
	CA-CFAR ($\alpha = 3.5$)	86.7 ± 29.8	5.2 ± 0.7	+0.86	+21.20	+20.35 ± 5.38
	Median–MAD	86.7 ± 29.8	6.4 ± 1.3	+0.86	+21.20	+20.35 ± 5.38
3 HP (7.2 N·m)	Static (0.055)	0.0 ± 0.0	0.0 ± 0.0	−2.06	+24.60	+26.66 ± 1.56
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	6.1 ± 1.0	−2.06	+24.60	+26.66 ± 1.56
	Median–MAD	100.0 ± 0.0	7.9 ± 0.7	−2.06	+24.60	+26.66 ± 1.56

keeps each conjugate pair in its own biquad, so a rounding error in one section stays there. The largest pole radius of the designed filter is $r_{\max} = 0.9626$; the region of convergence $|z| > 0.9626$ contains the unit circle, so the cascade stays BIBO stable.

2) Quantization noise and SQNR. Over the filtered passband the signal-to-quantization-noise ratio is 29.76 dB. For $B = 16$ bits the theoretical limit is

$$\text{SQNR}_{\text{theoretical}} \approx 6.02B + 1.76 \text{ dB} = 98.08 \text{ dB}, \qquad (38)$$

so the implementation lands about 68 dB below the ideal figure. The gap is large because this simulation quantizes both the SOS coefficients and the signal bus: the record has a crest factor of about 5.3, so the Q1.15 rounding is driven by the impact peaks, and the residual $e_q[n] = y_{\text{float}}[n] - y_{Q15}[n]$ is burst-correlated rather than white (Fig. 4(b)). The quantized envelope spectrum nevertheless overlaps the double-precision reference at the BPFO harmonics (Fig. 4(a)), so the harmonic decision is unchanged.

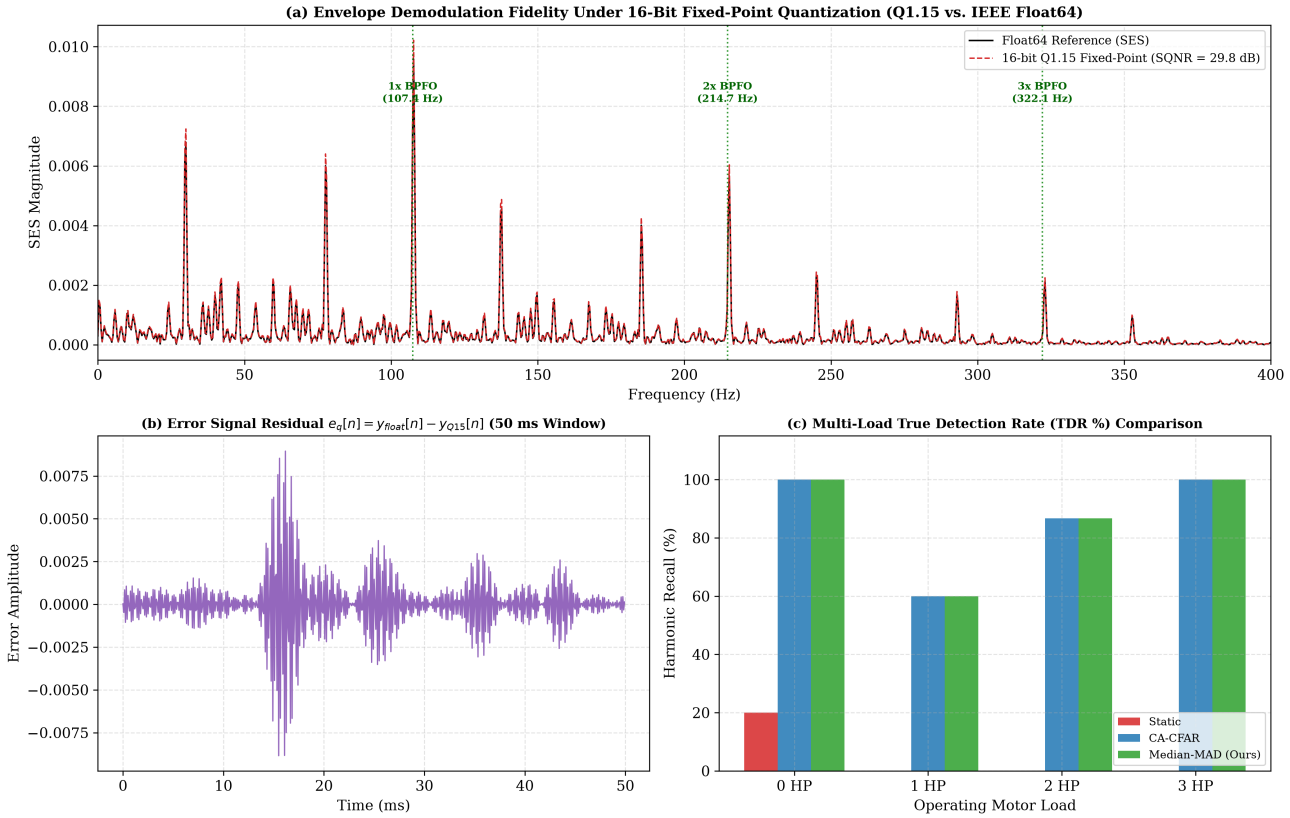


Fig. 4. Finite-word-length evaluation on the 0 HP outer-race record: (a) calibrated SES under Q1.15 fixed-point filtering against the float64 reference, (b) quantization residual $e_q[n]$ over a 50 ms window, and (c) detector TDR across the four CWRU loads.

V. Conclusion and Hardware Feasibility

A. Summary of Diagnostic Achievements

Four results came out of the end-to-end run on the CWRU and MFPT records, which starts at the raw accelerometer channel and ends at a thresholded envelope spectrum.

1) The Fast Kurtogram found the resonance band by itself: $f_c = 5181.8$ Hz with 545.5 Hz of bandwidth on the 0 HP outer-race record, and no manual tuning of the filter.
2) CA-CFAR and Median–MAD both recovered the BPFO harmonic comb at every load, with mean detection rates of 100, 60, 86.7 and 100% from no load to full load.
3) Zero-phase Butterworth SOS filtering, followed by coherent-gain calibrated Hilbert envelope analysis, gave between 19.2 and 26.7 dB of spectral SNR improvement across the load sweep.
4) At the design value $\gamma = 4.5$ the Median–MAD rule matched CA-CFAR's detection rate but crossed more non-harmonic bins (6.4–10.3% against 5.2–7.0%); the fixed 0.055 threshold did not transfer across loads.

B. Embedded Hardware Feasibility and Real-Time Budget

The same pipeline was timed and re-run in fixed point, to check its fit on an edge device.

1) *Memory.* A 2.0 s block ($N = 24\,000$ samples) at 16 bits per sample needs 48 kB of SRAM. That fits an ARM Cortex-M4 or M7 part such as the STM32F4/H7, which carries 512 kB.
2) *Execution latency.* The interpreted Python (CPython/Colab) pipeline takes between 0.35 and 0.45 s for a 2.0 s window, so the duty cycle is

$$\eta = \frac{T_{\text{compute}}}{T_{\text{acquire}}} \approx \frac{0.35\text{–}0.45 \text{ s}}{2.0 \text{ s}} = 17\text{–}23\%. \qquad (39)$$

Even without compilation the chain is 4.4–5.7 times faster than real time, and the processor idles for more than three quarters of the timeline. The figure is a software timing, not an embedded benchmark. The MFPT records, sampled at 48.828 kHz, take about 0.71 s per 2.0 s segment.
3) *Fixed-point suitability.* At 29.76 dB the Q1.15 SQNR is far below the 98.08 dB ideal, because the simulation quantizes both coefficients and the signal bus on a record with a crest factor near five. The harmonic decision survives (Section IV-F), so integer MAC units can run the filter, while the envelope FFT stage keeps a wider internal word length.

Appendix A
CWRU Baseline Records

These figures come from the baseline analysis of Section III: the preprocessing chain, the static spectrum and the window comparison, all taken before any fault detection runs.

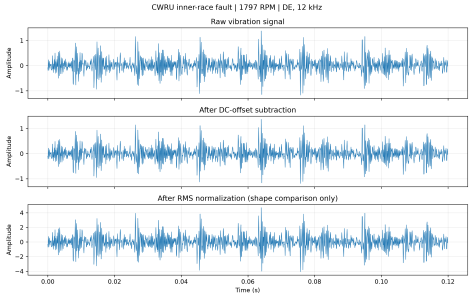


Fig. 5. CWRU inner-race fault: time-domain preprocessing (raw, DC-removed and RMS-normalized signals).

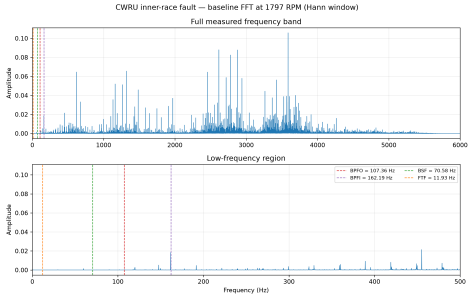


Fig. 6. CWRU inner-race fault: baseline FFT spectrum at 1797 rpm.

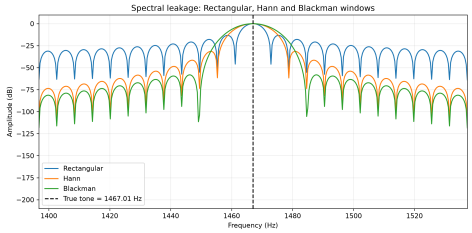


Fig. 7. Controlled off-bin single-tone comparison of the rectangular, Hann and Blackman windows.

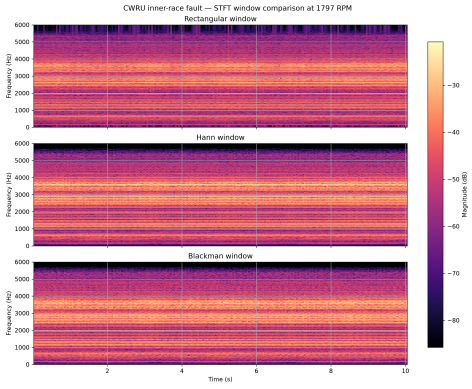


Fig. 8. CWRU inner-race fault: STFT comparison using the rectangular, Hann and Blackman windows.

Appendix B
MFPT Baseline Records

The second set follows the MFPT bearing through its load sweep, before and after the resonance band is selected.

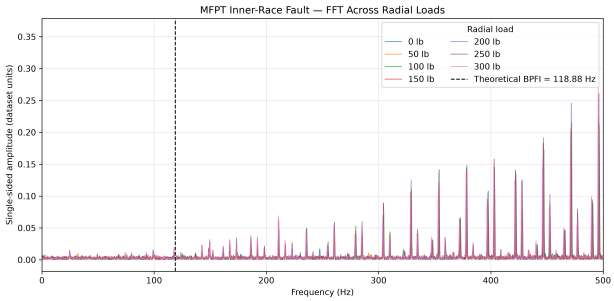


Fig. 9. MFPT inner-race fault: FFT across radial loads.

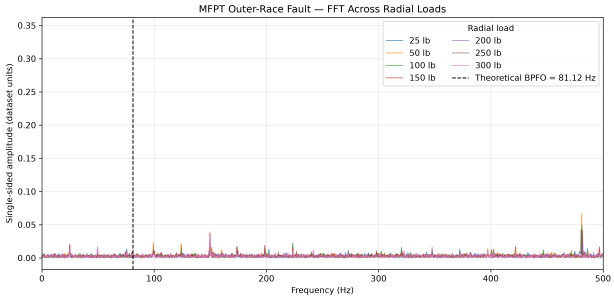


Fig. 10. MFPT outer-race fault: FFT across radial loads.

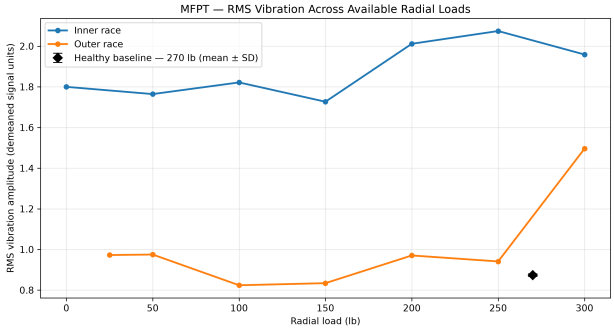


Fig. 11. MFPT RMS vibration amplitude across radial loads.

Acknowledgment

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Author Contributions

Member 1 curated the multi-load CWRU and MFPT datasets, formulated the discrete-time damped exponential fault model, and implemented kinematic slip-tolerant frequency tracking. **Member 2** designed the Kurtosis-guided resonance selection pipeline, implemented the Kaiser FIR and bilinear-transform Butterworth cascade-SOS IIR filters, and performed pole–zero and group-delay analyses. **Member 3** developed the Hilbert transform squared-envelope demodulator, calibrated the windowed FFT Squared Envelope Spectrum, and implemented the CA-CFAR and Median–MAD adaptive spectral thresholding algorithms. **Member 4** conducted the multi-load quantitative benchmarking, evaluated fixed-point quantization

feasibility, and led the integration of the manuscript. All authors reviewed and approved the final manuscript.

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